

Exercise 4

1. Syntax `SELECT order_date, order_id, customer_id,
DAYNAME('order_date') AS day_name
FROM orders-table;`

Orders-table

order_id	customer_id	order_date	day_name
1001	101	2026-05-01	Friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	Monday
1005	105	2026-05-05	Tuesday

2. Syntax `SELECT signup_date, customer_id, customer_name,
MONTHNAME('signup_date') AS signup-month-name
FROM customer-signups;`

Customer-signups

customer_id	customer_name	signup_date	signup-month-name
201	John	2026-01-15	January
202	Mary	2026-02-20	February
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-30	May

3. Syntax `SELECT sale_id, product_name, sale_date,
MONTH('sale_date') AS sale-month
FROM sales-table;`

Sales-table

Sales_id	product_name	sale_date	sale-month
S001	Laptop	2026-01-10	01
S002	Mouse	2026-02-15	02
S003	Keyboard	2026-03-20	03
S004	Monitor	2026-04-25	04
S005	Desk	2026-05-30	05

4. Syntax ~~SELECT EXTRACT(YEAR FROM~~

~~SELECT transaction-id, customer-id, transaction-date,
EXTRACT(YEAR FROM 'transaction-date') AS transaction-year
FROM transaction-table~~

transaction-table

transaction-id	customer-id	transaction-date	transaction-year
T001	101	2024-12-15	2024
T002	102	2025-01-20	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

5 Syntax ~~SELECT delivery-id, customer-id, delivery-date, DAY('delivery-date') AS day-of-month FROM deliveries-table;~~

deliveries-table

delivery-id	customer-id	delivery-date	day-of-month
D001	101	2026-05-01	Friday
D002	102	2026-05-08	Friday
D003	103	2026-05-15	Friday
D004	104	2026-05-22	Friday
D005	105	2026-05-29	Friday

6. Syntax ~~SELECT employee-id, employee-name, department, CURRENT-DATE() AS today-date (CURRENTDATE(2026-06-20)) AS today-date FROM employees-table~~

employees-table

employee-id	employee-name	department	today-date
301	Mandi	Sales	2026-06-20
302	Brian	IT	2026-06-20
303	Lerato	Finance	2026-06-20
304	Sipho	HR	2026-06-20
305	Aisha	Marketing	2026-06-20

7. Syntax `SELECT order_id, customer_id, order_date_text,
TO_DATE('order_date_text', 'dd-mm-yyyy') As order_date
FROM online-orders-table;`

online-orders-table

order_id	customer_id	order_date_text	order_date
4001	101	2026-01-15	15-01-2026
4002	102	2026-02-20	20-02-2026
4003	103	2026-03-25	25-03-2026
4004	104	2026-04-10	10-04-2026
4005	105	2026-05-05	05-05-2026

8. Syntax `SELECT payment_id, customer_id, payment_date,
TO_CHAR(payment_date, 'yyyy-mm-dd') As formatted_payment_date
FROM payment-dates-table;`

payment-dates-table

payment_id	customer_id	payment_date	formatted_payment_date
P001	101	2026-01-05	2026-01-05
P002	102	2026-02-10	2026-02-10
P003	103	2026-03-15	2026-03-15
P004	104	2026-04-20	2026-04-20
P005	105	2026-05-25	2026-05-25

9. Syntax `SELECT customer_id, customername, last-purchase-date,
CURRENT_DATE ('2026-06-20'),
DATE DIFF ('last-purchase-date', '2026-06-20', 'last-purchase-date') AS
days-since-last-purchase
FROM customer-purchases`

customer-purchases-table

customer_id	customer-name	last-purchase-date	days-since-last-purchase
501	Peter John	2026-05-01	50
502	Mary	2026-05-10	40
503	Peter	2025-05-15	35
504	Sarah	2025-05-20	31
505	Thabo	2025-05-25	26

10 Syntax `SELECT order_id, customer_id, order-date,
DATE_ADD('order-date', 7) AS expected-delivery-date
FROM shipping-orders-table`

shipping-orders-table

order_id	customer_id	order-date	expected-delivery-date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

11. Syntax `SELECT booking_id, customer_id, booking-date,
YEAR('booking-date') AS booking-year
MONTH('booking-date') AS booking-month
DAY('booking-date') AS booking-day
FROM bookings-table`

bookings-table

booking-id	customer-id	booking-date	booking-year	booking-month	booking-day
B001	101	2026-01-12	2026	01	12
B002	102	2026-02-18	2026	02	18
B003	103	2026-03-22	2026	03	22
B004	104	2026-04-09	2026	04	09
B005	105	2026-05-27	2026	05	27

12. Syntax SELECT order-id, customer-id, order-date, amount
 YEAR('order-date') As order-year
 FROM yearly-orders-table
 WHERE order-date = 2026;

yearly-orders-table

order-id	customer-id	order-date	order-year	amount
7004	104	2026-02-05	2026	1500
7005	105	2026-05-25	2026	2000

13. Syntax SELECT order-id, customer-id, order-date, amount,
 MONTH('order-date') As order-month
 FROM monthly-orders-table
 WHERE order-date = March 03;

monthly-orders-table

order-id	customer-id	order-date	order-month	amount
8003	103	2026-03-10	03	800
8004	104	2026-03-25	03	1500

14. Syntax: SELECT subscription_id, customer_id, start_date,
LAST_DAY(start_date) As month_end_date
FROM subscriptions;

subscriptions-table

subscription_id	customer_id	start_date	month_end_date
SUB001	101	2026-01-10	31
SUB002	102	2026-02-15	28
SUB003	103	2026-03-20	31
SUB004	104	2026-04-25	30
SUB005	105	2026-05-30	31

15 Syntax: SELECT send_id, customer_id, send_date,
DATE_TRUNC('month', send_date) As month_start_date
FROM campaign_sends;

campaign_sends

send_id	customer_id	send_date	month_start_date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-18	2026-02-01
C003	103	2026-03-12	2026-03-01
C004	104	2026-04-09	2026-04-01
C005	105	2026-05-27	2026-05-01

16. Syntax: SELECT invoice_id, customer_id, invoice_date,
TO_CHAR(invoice_date, 'mmmm-yyyy') As invoice-month-year
FROM invoice_dates;

invoice_dates

invoice_id	customer_id	invoice_date	invoice-month-year
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	February 2026
INV003	103	2026-03-15	March 2026
INV004	104	2026-04-20	May 2026
INV005	105	2026-05-25	June 2026

17 Syntax: SELECT customer_id, customer_name, date_of_birth
 CURRENT_DATE ('2026-06-20')
 DATEDIFF ('2026-06-20', 'date_of_birth') AS customer_age
 FROM customer_birthdays;

customer_birthdays

customer_id	customer_name	date_of_birth	customer_age
901	John	1998-05-10	28
902	Mary	1990-08-20	36
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	41
905	Thabo	2000-07-30	26

18 Syntax: SELECT order_id, customer_id, order_date,
 DAYNAME (order_date) AS day_name

CASE

WHERE day_name = 'Saturday AND Sunday' THEN ^(weekend) ~~WEEKEND~~

WHERE day_name = "All others" THEN 'week days'

END AS day_type

FROM weekend_orders

weekend_orders table

order_id	customer_id	order_date	day_name	day_type
9001	101	2026-05-01	Friday	weekday
9002	102	2026-05-02	Saturday	weekend
9003	103	2026-05-03	Sunday	weekend
9004	104	2026-05-04	Monday	weekday
9005	105	2026-05-05	Tuesday	weekday

19. Syntax: SELECT transaction_id, customer_id, transaction_date, amount,
 QUARTER(transaction_date) As transaction_quarter
 FROM quarterly_transactions

quarterly_transactions table

transaction_id	customer_id	transaction_date	transaction_quarter	amount
Q001	101	2026-01-15	03	500
Q002	102	2026-03-20	03	1200
Q003	103	2026-04-10	06	800
Q004	104	2026-07-05	09	1500
Q005	105	2026-10-25	12	2000

20. Syntax: SELECT order_id, customer_id, order_date, amount,
 CURRENT_DATE(2026-06-21)
 DATEDIFF('2026-06-21', order_date) As days_since_order
 FROM recent_orders_table
 WHERE days_since_orders > 30;

recent_orders_table

order_id	customer_id	order_date	days_since_order	amount
R001	101	2026-04-01	82	500
R002	102	2026-04-15	67	1200
R003	103	2026-05-01	52	800
R004	104	2026-05-25	43	1500
R005	105			

B Syntax: SELECT customer_id, customer_name, last_purchase_date,
 CURRENT_DATE(2026-06-21)
 DATEDIFF(2026-06-21, last_purchase_date) As days_since_last_purchase
 CASE WHEN days_since_last_purchase ≤ 30 THEN 'Active customer'
 WHEN days_since_last_purchase BETWEEN 31 AND 90 THEN 'At risk customer'
 ELSE 'Inactive customer' END AS customer_status
 FROM customer_recoveryency

Customer_reclency

customer_id	customer_name	last_purchase_date	days_since_last_purchase	customer_status
1001	John	2026-05-25	52	At risk customer
1002	Mary	2026-05-10	43	At risk customer
1003	Peter	2026-04-01	82	At risk customer
1004	Sarah	2026-02-15	127	Inactive
1005	Thabo	2025-12-20	184	Inactive